

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import scipy.stats as stats
from scipy.stats import ttest_ind
from scipy.stats import f_oneway
```

Step 1: Understand Your Data

Load and preview the dataset.

```
US_Customer_Insights_Dataset = pd.read_csv("C:\\Users\\Admin\\Desktop\\Stastics\\Stastics project\\Dataset\\US_Customer_Insights_Dataset.csv")
US_Customer_Insights_Dataset
```

	CustomerID	Name	State	Education	Gender	Age	Married	NumPets	JoinDate	TransactionDate	MonthlySpend	DaysSinceLastInteraction
0	CUST10319	Scott Perez	Florida	High School	Non-Binary	47	Yes	1	9/19/21	9/2/24	1281.74	332
1	CUST10695	Jennifer Burton	Washington	Master	Male	72	Yes	0	4/5/24	6/2/24	429.46	424
2	CUST10297	Michelle Rogers	Arizona	Master	Female	40	Yes	2	7/24/24	2/28/25	510.34	153
3	CUST10103	Brooke Hendricks	Texas	Master	Male	27	Yes	0	8/12/23	3/29/25	396.47	124
4	CUST10219	Karen Johns	Texas	High School	Female	28	Yes	1	12/6/21	7/24/22	139.68	1103
...	...	...	...	...	...	...	...	...	...	...	...	...
10670	CUST10833	Steven Burns	Georgia	PhD	Female	60	No	1	8/24/23	2/29/24	341.28	518
10671	CUST10620	Jesse Pratt	Texas	Master	Male	64	No	0	4/13/23	12/31/24	468.04	212
10672	CUST10449	John Lloyd	Arizona	Master	Non-Binary	31	Yes	0	7/3/22	9/21/23	259.94	679
10673	CUST10020	Christopher Sparks	Florida	Bachelor	Female	31	No	0	9/19/23	12/29/23	494.17	580
10674	CUST10267	Melissa Marshall	Arizona	Associate	Non-Binary	57	Yes	1	4/3/23	12/1/23	153.12	608

10675 rows × 12 columns

```
US_Customer_Insights_Dataset.head(10)
```

	CustomerID	Name	State	Education	Gender	Age	Married	NumPets	JoinDate	TransactionDate	MonthlySpend	DaysSinceLastInteraction
0	CUST10319	Scott Perez	Florida	High School	Non-Binary	47	Yes	1	9/19/21	9/2/24	1281.74	332
1	CUST10695	Jennifer Burton	Washington	Master	Male	72	Yes	0	4/5/24	6/2/24	429.46	424
2	CUST10297	Michelle Rogers	Arizona	Master	Female	40	Yes	2	7/24/24	2/28/25	510.34	153
3	CUST10103	Brooke Hendricks	Texas	Master	Male	27	Yes	0	8/12/23	3/29/25	396.47	124
4	CUST10219	Karen Johns	Texas	High School	Female	28	Yes	1	12/6/21	7/24/22	139.68	1103
5	CUST10841	Stephen Cooke	Texas	PhD	Female	55	Yes	2	5/25/22	4/4/23	423.45	849
6	CUST10263	Rodney James	Ohio	High School	Non-Binary	61	No	0	11/20/23	3/26/24	402.96	492
7	CUST10117	Jeffrey Miles	Arizona	High School	Male	21	Yes	2	9/28/20	7/9/23	175.81	753
8	CUST10611	John Walsh	New York	Master	Male	74	No	2	5/15/21	7/30/22	320.48	1097
9	CUST10466	James Pope	New York	Master	Female	50	No	1	7/28/21	1/28/23	151.49	915

US_Customer_Insights_Dataset.tail(10)

	CustomerID	Name	State	Education	Gender	Age	Married	NumPets	JoinDate	TransactionDate	MonthlySpend	DaysSinceLastInteraction
10665	CUST10898	Heather Chavez	New York	Associate	Non-Binary	21	Yes	2	4/21/24	12/25/24	461.84	218
10666	CUST10564	Mary Shaw	Washington	Master	Female	41	No	0	7/13/24	11/11/24	483.33	262
10667	CUST10679	Ann Rose	Arizona	Associate	Male	54	Yes	0	10/4/22	7/17/25	839.65	14
10668	CUST10965	Kristin Dillon	Illinois	Master	Male	40	Yes	2	7/29/22	11/29/24	171.54	244
10669	CUST10596	Gary James	Georgia	Associate	Non-Binary	39	Yes	1	3/12/24	1/24/25	331.24	188
10670	CUST10833	Steven Burns	Georgia	PhD	Female	60	No	1	8/24/23	2/29/24	341.28	518
10671	CUST10620	Jesse Pratt	Texas	Master	Male	64	No	0	4/13/23	12/31/24	468.04	212
10672	CUST10449	John Lloyd	Arizona	Master	Non-Binary	31	Yes	0	7/3/22	9/21/23	259.94	679
10673	CUST10020	Christopher Sparks	Florida	Bachelor	Female	31	No	0	9/19/23	12/29/23	494.17	580
10674	CUST10267	Melissa Marshall	Arizona	Associate	Non-Binary	57	Yes	1	4/3/23	12/1/23	153.12	608

US_Customer_Insights_Dataset.shape

(10675, 12)

```
US_Customer_Insights_Dataset.info
```

```
<bound method DataFrame.info of
0      CUST10319      Scott Perez      Florida      High School      Non-Binary
1      CUST10695      Jennifer Burton      Washington      Master      Male
2      CUST10297      Michelle Rogers      Arizona      Master      Female
3      CUST10103      Brooke Hendricks      Texas      Master      Male
4      CUST10219      Karen Johns      Texas      High School      Female
...
10670      CUST10833      Steven Burns      Georgia      PhD      Female
10671      CUST10620      Jesse Pratt      Texas      Master      Male
10672      CUST10449      John Lloyd      Arizona      Master      Non-Binary
10673      CUST10020      Christopher Sparks      Florida      Bachelor      Female
10674      CUST10267      Melissa Marshall      Arizona      Associate      Non-Binary
```

```
      Age      Married      NumPets      JoinDate      TransactionDate      MonthlySpend      \
0      47      Yes      1      9/19/21      9/2/24      1281.74
1      72      Yes      0      4/5/24      6/2/24      429.46
2      40      Yes      2      7/24/24      2/28/25      510.34
3      27      Yes      0      8/12/23      3/29/25      396.47
4      28      Yes      1      12/6/21      7/24/22      139.68
...
10670      60      No      1      8/24/23      2/29/24      341.28
10671      64      No      0      4/13/23      12/31/24      468.04
10672      31      Yes      0      7/3/22      9/21/23      259.94
10673      31      No      0      9/19/23      12/29/23      494.17
10674      57      Yes      1      4/3/23      12/1/23      153.12
```

```
      DaysSinceLastInteraction
0      332
1      424
2      153
3      124
4      1103
...
10670      518
10671      212
10672      679
10673      580
10674      608
```

```
[10675 rows x 12 columns]>
```

Check data types, unique values, and presence of nulls.

```
US_Customer_Insights_Dataset.columns
```

```
Index(['CustomerID', 'Name', 'State', 'Education', 'Gender', 'Age', 'Married',  
      'NumPets', 'JoinDate', 'TransactionDate', 'MonthlySpend',  
      'DaysSinceLastInteraction'],  
      dtype='str')
```

```
US_Customer_Insights_Dataset.dtypes
```

```
CustomerID      str  
Name            str  
State           str  
Education       str  
Gender          str  
Age             int64  
Married         str  
NumPets         int64  
JoinDate        str  
TransactionDate str  
MonthlySpend    float64  
DaysSinceLastInteraction int64  
dtype: object
```

```
US_Customer_Insights_Dataset['JoinDate'] = pd.to_datetime(US_Customer_Insights_Dataset['JoinDate'])  
US_Customer_Insights_Dataset.dtypes
```

```
CustomerID      str  
Name            str  
State           str  
Education       str  
Gender          str  
Age             int64  
Married         str  
NumPets         int64  
JoinDate        datetime64[us]  
TransactionDate datetime64[us]  
MonthlySpend    float64  
DaysSinceLastInteraction int64  
dtype: object
```

```
US_Customer_Insights_Dataset['TransactionDate'] = pd.to_datetime(US_Customer_Insights_Dataset['TransactionDate'])
US_Customer_Insights_Dataset.dtypes
```

```
CustomerID          str
Name                str
State              str
Education           str
Gender             str
Age                int64
Married            str
NumPets            int64
JoinDate            datetime64[us]
TransactionDate     datetime64[us]
MonthlySpend        float64
DaysSinceLastInteraction int64
dtype: object
```

US_Customer_Insights_Dataset

	CustomerID	Name	State	Education	Gender	Age	Married	NumPets	JoinDate	TransactionDate	MonthlySpend	DaysSinceLastInteraction
0	CUST10319	Scott Perez	Florida	High School	Non-Binary	47	Yes	1	2021-09-19	2024-09-02	1281.74	332
1	CUST10695	Jennifer Burton	Washington	Master	Male	72	Yes	0	2024-04-05	2024-06-02	429.46	424
2	CUST10297	Michelle Rogers	Arizona	Master	Female	40	Yes	2	2024-07-24	2025-02-28	510.34	153
3	CUST10103	Brooke Hendricks	Texas	Master	Male	27	Yes	0	2023-08-12	2025-03-29	396.47	124
4	CUST10219	Karen Johns	Texas	High School	Female	28	Yes	1	2021-12-06	2022-07-24	139.68	1103
...	...	...	...	...	...	...	...	...	...	...	...	...
10670	CUST10833	Steven Burns	Georgia	PhD	Female	60	No	1	2023-08-24	2024-02-29	341.28	518
10671	CUST10620	Jesse Pratt	Texas	Master	Male	64	No	0	2023-04-13	2024-12-31	468.04	212
10672	CUST10449	John Lloyd	Arizona	Master	Non-Binary	31	Yes	0	2022-07-03	2023-09-21	259.94	679
10673	CUST10020	Christopher Sparks	Florida	Bachelor	Female	31	No	0	2023-09-19	2023-12-29	494.17	580
10674	CUST10267	Melissa Marshall	Arizona	Associate	Non-Binary	57	Yes	1	2023-04-03	2023-12-01	153.12	608

10675 rows × 12 columns

```
US_Customer_Insights_Dataset.duplicated().sum()
```

```
np.int64(0)
```

```
US_Customer_Insights_Dataset.isnull().sum()
```

```
CustomerID      0
Name            0
State           0
Education       0
Gender          0
Age            0
Married         0
NumPets         0
JoinDate        0
TransactionDate 0
MonthlySpend    0
DaysSinceLastInteraction 0
dtype: int64
```

```
US_Customer_Insights_Dataset.isna().sum()
```

```
CustomerID      0
Name            0
State           0
Education       0
Gender          0
Age            0
Married         0
NumPets         0
JoinDate        0
TransactionDate 0
MonthlySpend    0
DaysSinceLastInteraction 0
dtype: int64
```

```
US_Customer_Insights_Dataset['CustomerID'].unique()
```

```
<StringArray>  
['CUST10319', 'CUST10695', 'CUST10297', 'CUST10103', 'CUST10219', 'CUST10841',  
 'CUST10263', 'CUST10117', 'CUST10611', 'CUST10466',  
 ...  
 'CUST10460', 'CUST10669', 'CUST10415', 'CUST10164', 'CUST10166', 'CUST10346',  
 'CUST10201', 'CUST10271', 'CUST10312', 'CUST10268']  
Length: 1000, dtype: str
```

```
US_Customer_Insights_Dataset['Name'].unique()
```

```
<StringArray>  
[ 'Scott Perez',    'Jennifer Burton',    'Michelle Rogers',  
  'Brooke Hendricks', 'Karen Johns',    'Stephen Cooke',  
  'Rodney James',    'Jeffrey Miles',    'John Walsh',  
  'James Pope',  
 ...  
  'Vanessa Smith',    'Jeff Diaz',    'Vincent Spencer',  
  'Patricia Jacobson', 'Ashley Burns', 'Maria Johnson',  
  'Debra Chavez',    'Andrea Williams', 'Erin Norman',  
  'Michelle Martinez']  
Length: 990, dtype: str
```

```
US_Customer_Insights_Dataset['State'].unique()
```

```
<StringArray>  
[ 'Florida', 'Washington', 'Arizona', 'Texas', 'Ohio',  
  'New York', 'Illinois', 'Georgia', 'California', 'Colorado']  
Length: 10, dtype: str
```

```
US_Customer_Insights_Dataset['Education'].unique()
```

```
<StringArray>  
['High School', 'Master', 'PhD', 'Bachelor', 'Associate']  
Length: 5, dtype: str
```

```
US_Customer_Insights_Dataset['Gender'].unique()
```

```
<StringArray>  
['Non-Binary', 'Male', 'Female']  
Length: 3, dtype: str
```

```
US_Customer_Insights_Dataset['Age'].unique()
```

```
array([47, 72, 40, 27, 28, 55, 61, 21, 74, 50, 49, 30, 25, 78, 39, 26, 51,  
       75, 38, 34, 68, 73, 37, 71, 43, 22, 79, 20, 52, 69, 18, 32, 58, 36,  
       53, 46, 45, 44, 41, 29, 35, 56, 80, 62, 65, 54, 77, 67, 60, 70, 31,  
       33, 64, 76, 59, 57, 23, 42, 63, 19, 66, 24, 48])
```

```
US_Customer_Insights_Dataset['Married'].unique()
```

```
<StringArray>  
['Yes', 'No']  
Length: 2, dtype: str
```

```
US_Customer_Insights_Dataset['NumPets'].unique()
```

```
array([1, 0, 2, 3, 4])
```

```
US_Customer_Insights_Dataset['JoinDate'].unique()
```

```
<DatetimeArray>  
['2021-09-19 00:00:00', '2024-04-05 00:00:00', '2024-07-24 00:00:00',  
 '2023-08-12 00:00:00', '2021-12-06 00:00:00', '2022-05-25 00:00:00',  
 '2023-11-20 00:00:00', '2020-09-28 00:00:00', '2021-05-15 00:00:00',  
 '2021-07-28 00:00:00',  
 ...  
 '2022-08-04 00:00:00', '2023-02-07 00:00:00', '2024-02-13 00:00:00',  
 '2021-03-18 00:00:00', '2021-10-22 00:00:00', '2024-04-04 00:00:00',  
 '2023-07-24 00:00:00', '2023-10-24 00:00:00', '2022-08-26 00:00:00',  
 '2023-08-07 00:00:00']  
Length: 731, dtype: datetime64[us]
```

```
US_Customer_Insights_Dataset['TransactionDate'].unique()
```

```
<DatetimeArray>  
['2024-09-02 00:00:00', '2024-06-02 00:00:00', '2025-02-28 00:00:00',  
 '2025-03-29 00:00:00', '2022-07-24 00:00:00', '2023-04-04 00:00:00',  
 '2024-03-26 00:00:00', '2023-07-09 00:00:00', '2022-07-30 00:00:00',  
 '2023-01-28 00:00:00',  
 ...  
 '2021-01-29 00:00:00', '2021-05-11 00:00:00', '2022-05-30 00:00:00',  
 '2020-11-29 00:00:00', '2021-10-14 00:00:00', '2021-03-09 00:00:00',  
 '2021-09-17 00:00:00', '2022-01-11 00:00:00', '2022-04-07 00:00:00',  
 '2021-06-04 00:00:00']  
Length: 1605, dtype: datetime64[us]
```

```
US_Customer_Insights_Dataset['MonthlySpend'].unique()
```

```
array([1281.74, 429.46, 510.34, ..., 259.94, 494.17, 153.12],  
      shape=(9843,))
```

```
US_Customer_Insights_Dataset['DaysSinceLastInteraction'].unique()
```

```
array([ 332, 424, 153, ..., 1297, 1211, 1518], shape=(1605,))
```

Understand which variables are categorical and which are numerical.

Categorical columns in the table are

- CustomerID
- Name
- State
- Education
- Gender
- Married

Numerical columns in the table are

- Age
- NumPets
- MonthlySpend
- DaysSinceLastInteraction

Step 2: Descriptive Statistics

Business Purpose: Describe your customer base — how old are they, how much do they spend, are they active?

```
US_Customer_Insights_Dataset['Age'].min()
```

```
np.int64(18)
```

```
US_Customer_Insights_Dataset['Age'].max()
```

```
np.int64(80)
```

```
US_Customer_Insights_Dataset['DaysSinceLastInteraction'].min()
```

```
np.int64(1)
```

```
US_Customer_Insights_Dataset['DaysSinceLastInteraction'].max()
```

```
np.int64(1791)
```

```
age_bins = [0, 25, 45, 65, 100]
age_labels = ['0-25', '25-45', '45-65', '65-100']
activity_threshold = 60

spending_and_activity = US_Customer_Insights_Dataset.groupby(
    pd.cut(US_Customer_Insights_Dataset['Age'], bins=age_bins, labels=age_labels),
    observed=False
).agg(
    average_spending=('MonthlySpend', 'mean'),
    activeCustomerCount=('DaysSinceLastInteraction', lambda x: (x <= activity_threshold).sum())
)

print(spending_and_activity)
```

	average_spending	activeCustomerCount
Age		
0-25	341.238840	98
25-45	328.841373	223
45-65	337.435316	234
65-100	323.309092	176

Compute:

- o Mean for Age

```
age_mean = US_Customer_Insights_Dataset['Age'].mean()
print(f"Mean of Age:{round(age_mean,2)}")
```

Mean of Age:49.47

o Median for Age

```
age_median = US_Customer_Insights_Dataset['Age'].median()
print(f"Median of Age:{round(age_median,2)}")
```

Median of Age:49.0

o std dev for Age

```
age_std = US_Customer_Insights_Dataset['Age'].std()
print(f"std of Age:{round(age_std,2)} years")
```

std of Age:18.22 years

o Mean for MonthlySpend

```
monthlySpend_mean = US_Customer_Insights_Dataset['MonthlySpend'].mean()
print(f"Mean of Monthly spend:{round(monthlySpend_mean,2)}")
```

Mean of Monthly spend:331.61

o Median for MonthlySpend

```
monthlySpend_median = US_Customer_Insights_Dataset['MonthlySpend'].median()
print(f"Median of Monthly spend:{round(monthlySpend_median,2)}")
```

Median of Monthly spend:282.11

o std dev MonthlySpend

```
monthlySpend_std = US_Customer_Insights_Dataset['MonthlySpend'].std()
print(f"std of Monthly spend:{round(monthlySpend_std,2)}")
```

std of Monthly spend:225.8

o Mean for DaysSinceLastInteraction

```
daysSinceLastInteraction_mean = US_Customer_Insights_Dataset['DaysSinceLastInteraction'].mean()
print(f"Mean of Days Since Last Interaction:{round(daysSinceLastInteraction_mean,2)}")
```

Mean of Days Since Last Interaction:538.47

o Median for DaysSinceLastInteraction

```
daysSinceLastInteraction_median = US_Customer_Insights_Dataset['DaysSinceLastInteraction'].median()
print(f"Median of Days Since Last Interaction:{round(daysSinceLastInteraction_median,2)}")
```

Median of Days Since Last Interaction:445.0

o std dev for DaysSinceLastInteraction

```
daysSinceLastInteraction_std = US_Customer_Insights_Dataset['DaysSinceLastInteraction'].std()
print(f"std of Days Since Last Interaction:{round(daysSinceLastInteraction_std,2)}")
```

```
std of Days Since Last Interaction:398.77
```

o Mode for categorical variables: Gender

```
US_Customer_Insights_Dataset['Gender'].value_counts()
```

```
Gender
Male      3791
Non-Binary 3471
Female    3413
Name: count, dtype: int64
```

```
Gender_mode = US_Customer_Insights_Dataset['Gender'].mode()[0]
print(f"Value of mode in Gender: {Gender_mode}")
```

```
Value of mode in Gender: Male
```

o Mode for categorical variables: Education

```
US_Customer_Insights_Dataset['Education'].value_counts()
```

```
Education
Master      2269
Associate   2153
Bachelor    2127
High School 2120
PhD         2006
Name: count, dtype: int64
```

```
Education_mode = US_Customer_Insights_Dataset['Education'].mode()[0]
print(f"Mode value for Education is {Education_mode}")
```

Mode value for Education is Master

- Mode for categorical variables: Married

```
US_Customer_Insights_Dataset['Married'].value_counts()
```

```
Married
No      5583
Yes     5092
Name: count, dtype: int64
```

```
Married_mode = US_Customer_Insights_Dataset['Married'].mode()[0]
print(f"Mode value for Married is: {Married_mode}")
```

Mode value for Married is: No

Step 3: Data Visualization

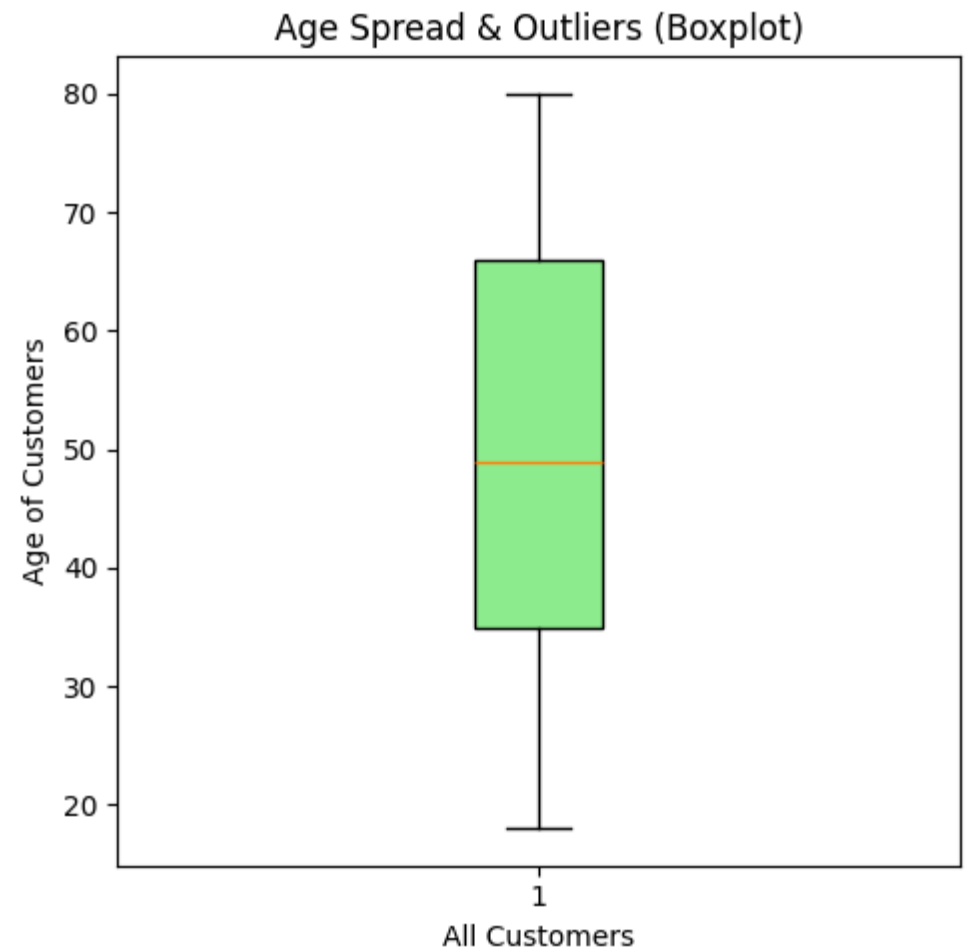
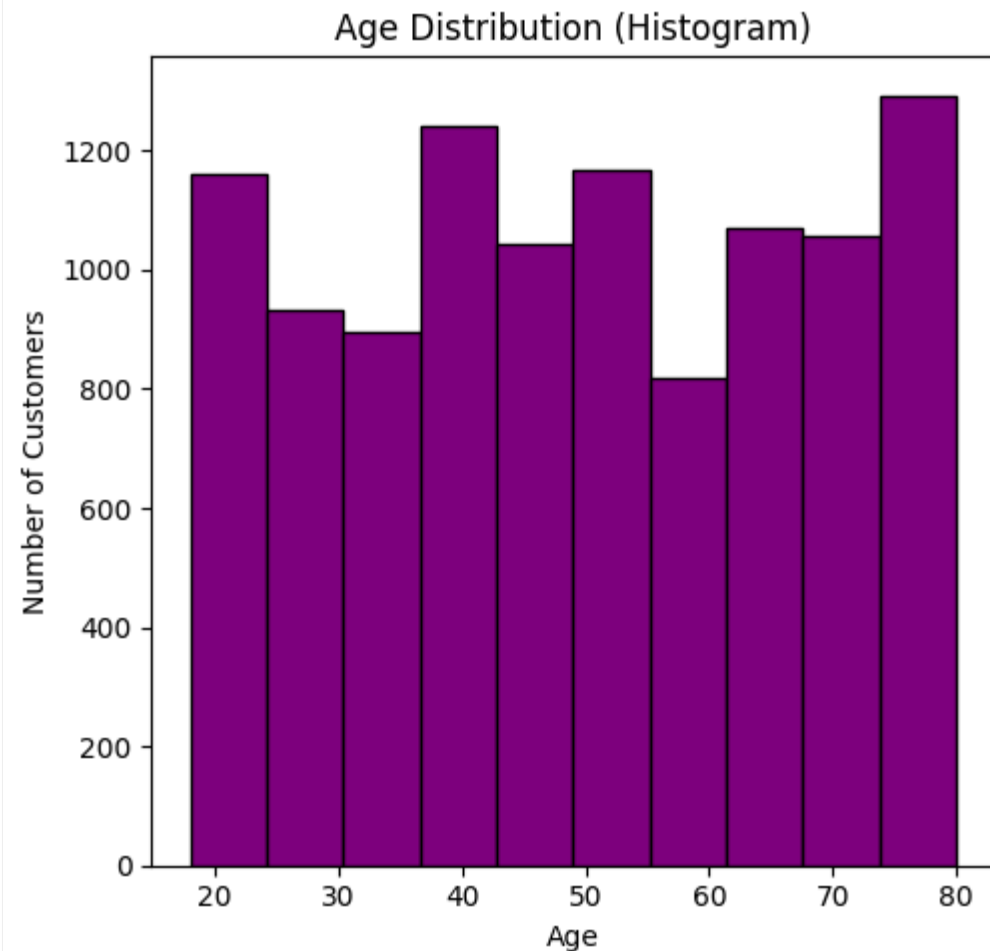
Business Purpose: Reveal patterns that numbers alone can't show.

Plot histograms and boxplots for Age

```

fig, ax = plt.subplots(1,2,figsize = (10,5))
ax[0].hist(US_Customer_Insights_Dataset['Age'],color = 'Purple', edgecolor = 'black')
ax[0].set_title('Age Distribution (Histogram)')
ax[0].set_xlabel('Age')
ax[0].set_ylabel('Number of Customers')
ax[1].boxplot(US_Customer_Insights_Dataset['Age'], patch_artist=True,boxprops=dict(facecolor='lightgreen'))
ax[1].set_xlabel('All Customers')
ax[1].set_ylabel('Age of Customers')
ax[1].set_title('Age Spread & Outliers (Boxplot)')
plt.tight_layout()
plt.show()

```

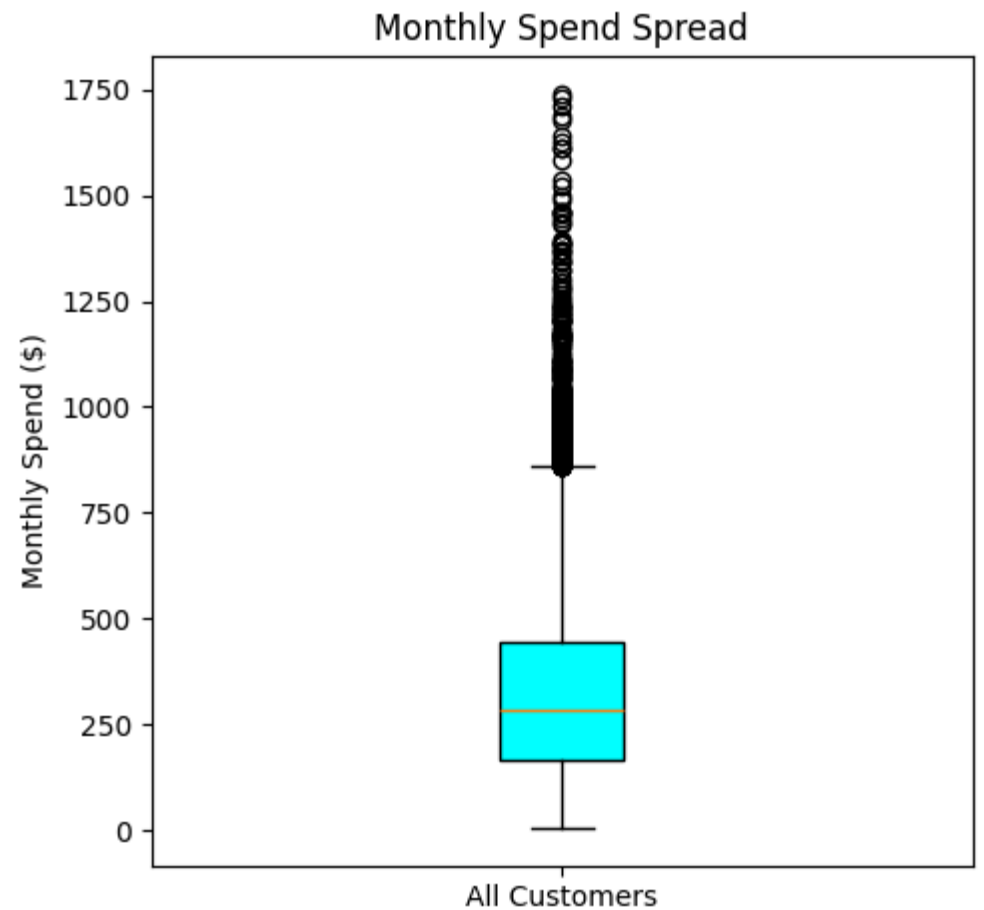
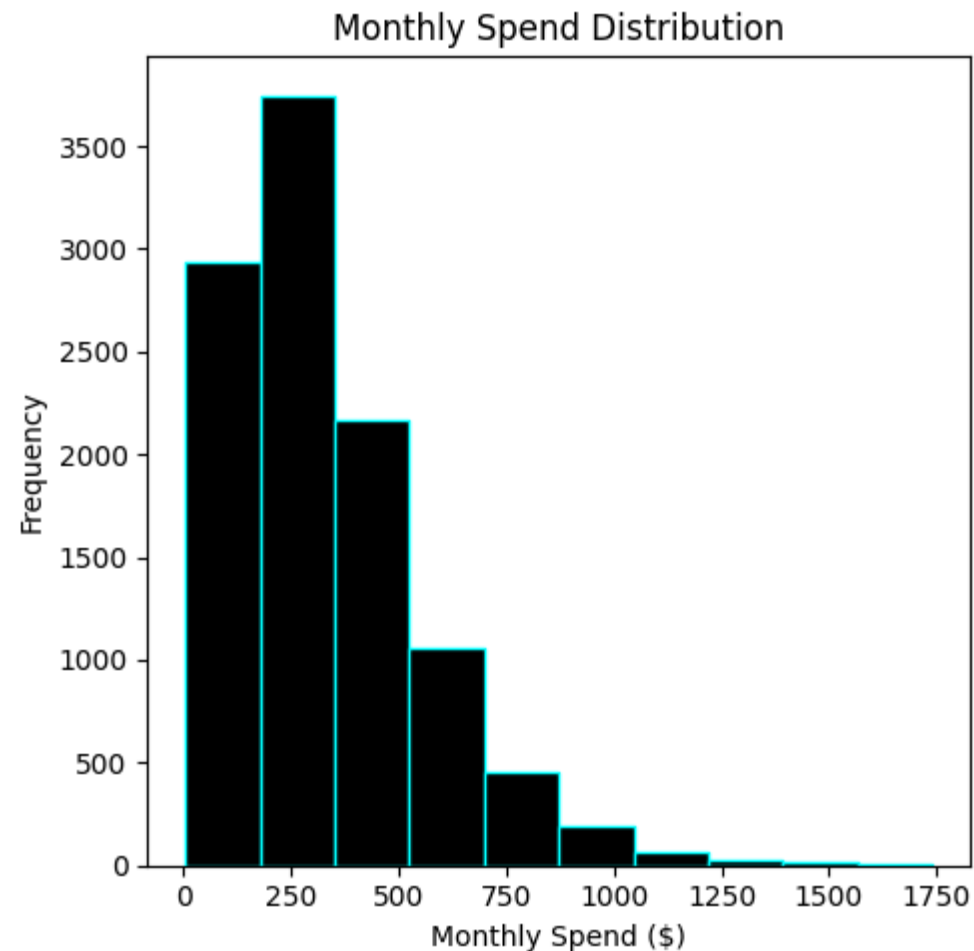


Plot histograms and boxplots for MonthlySpend

```

fig, ax = plt.subplots(1,2,figsize = (10,5))
ax[0].hist(US_Customer_Insights_Dataset['MonthlySpend'],color = 'Black', edgecolor = 'cyan')
ax[0].set_title('Monthly Spend Distribution')
ax[0].set_xlabel('Monthly Spend ($)')
ax[0].set_ylabel('Frequency')
ax[1].boxplot(US_Customer_Insights_Dataset['MonthlySpend'],vert=True, patch_artist=True,boxprops=dict(facecolor='Cyan'))
ax[1].set_title('Monthly Spend Spread')
ax[1].set_xticklabels(['All Customers'])
ax[1].set_ylabel('Monthly Spend ($)')
plt.tight_layout()
plt.show()

```



Create a bar chart for Gender, Education, State

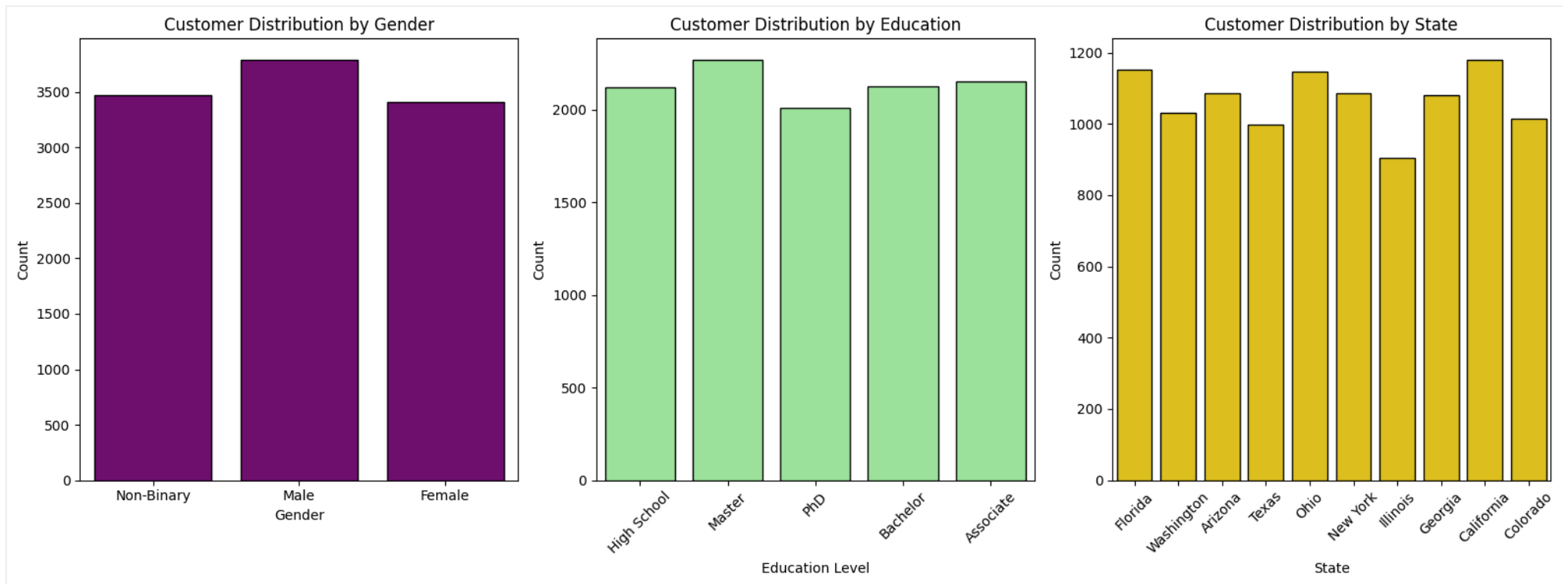
```
fig, ax = plt.subplots(1, 3, figsize=(16, 6))

sns.countplot(
    data=US_Customer_Insights_Dataset,
    x='Gender',
    color='purple',
    edgecolor='black',
    ax=ax[0]
)
ax[0].set_title('Customer Distribution by Gender')
ax[0].set_xlabel('Gender')
ax[0].set_ylabel('Count')

sns.countplot(
    data=US_Customer_Insights_Dataset,
    x='Education',
    color='lightgreen',
    edgecolor='black',
    ax=ax[1]
)
ax[1].set_title('Customer Distribution by Education')
ax[1].set_xlabel('Education Level')
ax[1].set_ylabel('Count')
ax[1].tick_params(axis='x', rotation=45)

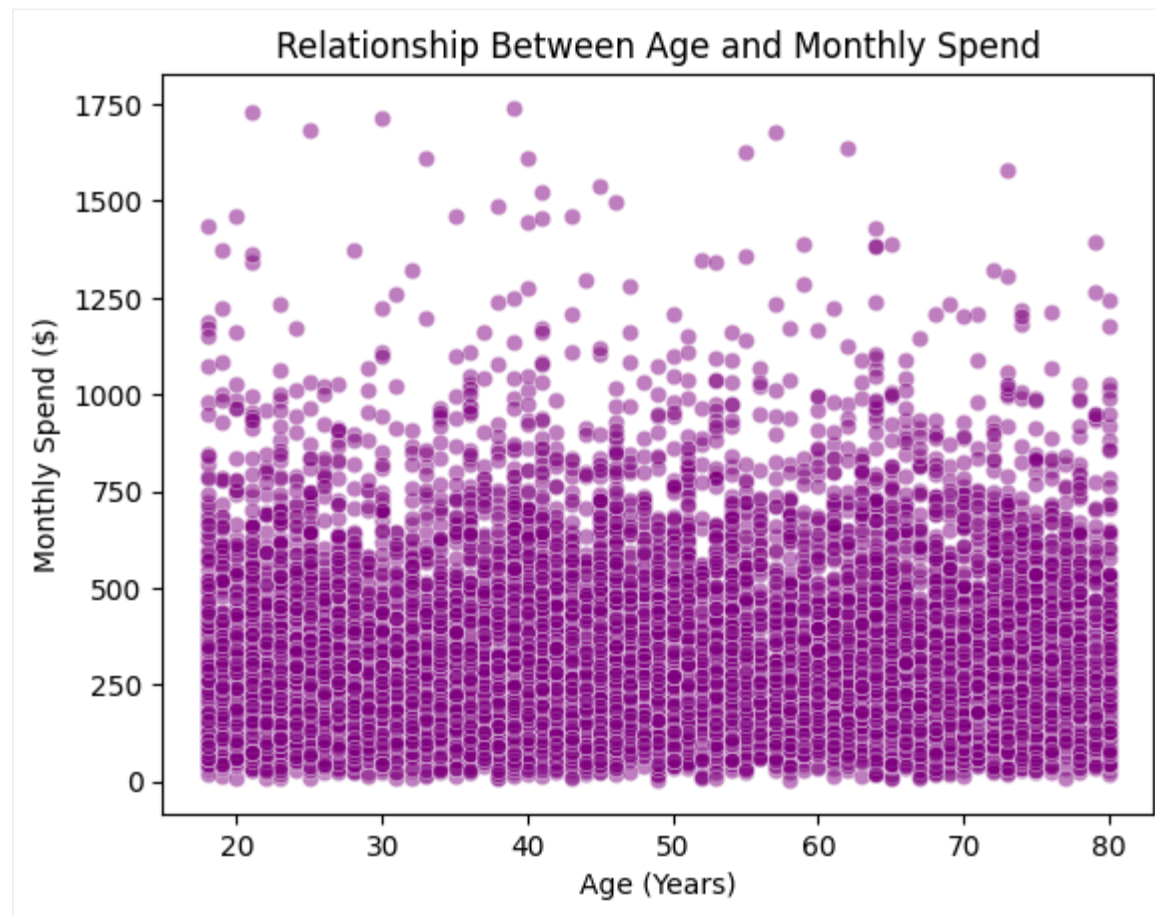
sns.countplot(
    data=US_Customer_Insights_Dataset,
    x='State',
    color='Gold',
    edgecolor='black',
    ax=ax[2]
)
ax[2].set_title('Customer Distribution by State')
ax[2].set_xlabel('State')
ax[2].set_ylabel('Count')
ax[2].tick_params(axis='x', rotation=45)

plt.tight_layout()
plt.show()
```



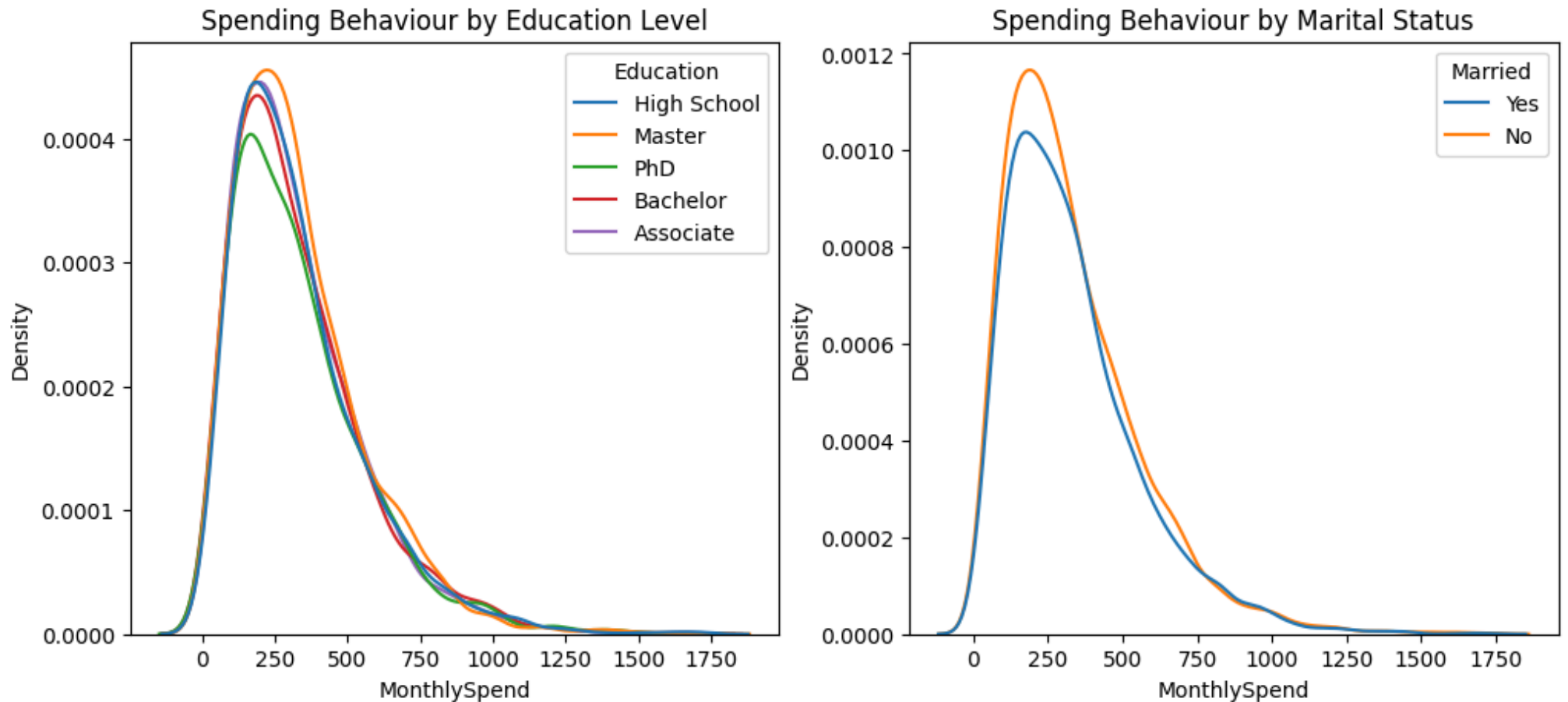
Scatterplot: Age vs MonthlySpend

```
sns.scatterplot(  
    x=US_Customer_Insights_Dataset['Age'],  
    y=US_Customer_Insights_Dataset['MonthlySpend'],  
    color = 'Purple',  
    alpha=0.5,  
)  
  
plt.title('Relationship Between Age and Monthly Spend')  
plt.xlabel('Age (Years)')  
plt.ylabel('Monthly Spend ($)')  
  
plt.show()
```



KDE: Spending behavior by education level or marital status

```
fig, ax = plt.subplots(1,2, figsize = (12,5))
sns.kdeplot(x = US_Customer_Insights_Dataset['MonthlySpend'], hue = US_Customer_Insights_Dataset['Education'],ax = ax[0])
ax[0].set_title('Spending Behaviour by Education Level')
sns.kdeplot(x = US_Customer_Insights_Dataset['MonthlySpend'], hue = US_Customer_Insights_Dataset['Married'],ax = ax[1])
ax[1].set_title('Spending Behaviour by Marital Status')
plt.show()
```



Step 4: Bivariate Analysis

Business Purpose: Check how customer attributes relate to one another.

Correlation matrix (numeric variables)

```
numeric_corr_matrix = US_Customer_Insights_Dataset.corr(numeric_only=True)

styled_matrix = numeric_corr_matrix.style.background_gradient(
    cmap="coolwarm", vmin=-1, vmax=1
).format(precision=2)

styled_matrix
```

	Age	NumPets	MonthlySpend	DaysSinceLastInteraction
Age	1.00	-0.02	-0.01	-0.00
NumPets	-0.02	1.00	0.02	-0.06
MonthlySpend	-0.01	0.02	1.00	0.01
DaysSinceLastInteraction	-0.00	-0.06	0.01	1.00

Crosstab of Gender vs Married

```
gender_married_xtab = pd.crosstab(  
    index=US_Customer_Insights_Dataset["Gender"],  
    columns=US_Customer_Insights_Dataset["Married"],  
    margins=True,  
    margins_name="Total",  
)  
  
print(gender_married_xtab)
```

Married	No	Yes	Total
Gender			
Female	1797	1616	3413
Male	1892	1899	3791
Non-Binary	1894	1577	3471
Total	5583	5092	10675

Grouped stats: average MonthlySpend by State, Education, Gender

```

pivot_spend = US_Customer_Insights_Dataset.pivot_table(
    values="MonthlySpend",
    index="State",
    columns=["Education", "Gender"],
    aggfunc="mean",
).round(2)

styled_pivot = pivot_spend.style.background_gradient(
    cmap="YlGnBu", axis=1
).format("${:,.2f}")

styled_pivot

```

Education	Associate			Bachelor			High School			Master			PhD		
Gender	Female	Male	Non-Binary	Female	Male	Non-Binary	Female	Male	Non-Binary	Female	Male	Non-Binary	Female	Male	Non-Binary
State															
Arizona	\$329.19	\$360.35	\$316.10	\$330.91	\$344.25	\$323.22	\$294.89	\$368.46	\$339.85	\$331.03	\$348.04	\$352.54	\$357.33	\$368.07	\$326.18
California	\$335.23	\$329.09	\$313.98	\$354.63	\$356.49	\$353.81	\$313.25	\$350.76	\$340.64	\$349.04	\$335.82	\$315.83	\$364.87	\$366.91	\$312.21
Colorado	\$314.29	\$310.60	\$309.55	\$328.45	\$327.63	\$336.85	\$317.89	\$322.18	\$307.94	\$320.30	\$317.25	\$328.46	\$329.02	\$314.91	\$352.64
Florida	\$310.23	\$318.68	\$332.81	\$310.27	\$349.59	\$358.65	\$334.61	\$346.80	\$330.26	\$341.80	\$339.34	\$336.15	\$319.04	\$289.87	\$318.36
Georgia	\$313.48	\$333.35	\$336.12	\$368.34	\$306.72	\$367.66	\$329.22	\$235.44	\$315.79	\$310.40	\$326.34	\$342.21	\$318.11	\$408.35	\$279.40
Illinois	\$350.41	\$332.32	\$336.16	\$377.82	\$305.08	\$355.12	\$355.96	\$343.15	\$347.91	\$315.93	\$337.17	\$300.85	\$314.24	\$319.27	\$297.81
New York	\$371.77	\$318.74	\$310.08	\$370.71	\$303.41	\$283.99	\$372.32	\$320.67	\$320.88	\$314.01	\$316.39	\$391.41	\$361.26	\$346.51	\$331.76
Ohio	\$316.18	\$359.24	\$359.58	\$318.36	\$308.77	\$305.13	\$329.12	\$375.85	\$337.33	\$374.87	\$353.34	\$377.91	\$306.21	\$329.84	\$351.36
Texas	\$318.47	\$318.66	\$351.18	\$290.73	\$336.80	\$315.02	\$302.11	\$354.33	\$284.00	\$333.92	\$324.34	\$354.02	\$300.65	\$335.27	\$300.33
Washington	\$319.74	\$350.67	\$284.96	\$320.81	\$322.11	\$339.84	\$307.26	\$316.07	\$356.30	\$362.95	\$305.58	\$318.77	\$368.06	\$333.00	\$351.27

Step 5: Formulate Hypotheses

Business Purpose: Turn business questions into statistical tests.

1. Do males and females spend differently?

- **Statistical Test:** Independent t-test
- **Null Hypothesis H0:** The mean `MonthlySpend` for males is equal to the mean `MonthlySpend` for females. $\mu(\text{male}) = \mu(\text{female})$
- **Alternative Hypothesis H1:** The mean `MonthlySpend` for males is significantly different from females. $\mu(\text{male}) \neq \mu(\text{female})$

2. Does education level impact average monthly spend?

- **Statistical Test:** One-Way ANOVA (Analysis of Variance)
- **Null Hypothesis H0:** The mean `MonthlySpend` is the same across all education levels. ($\mu(\text{High School}) = \mu(\text{Bachelors}) = \mu(\text{Masters}) = \dots$)
- **Alternative Hypothesis H1:** At least one education level group has a significantly different mean `MonthlySpend` compared to the others.

3. Are older people less active?

- **Statistical Test:** Correlation Test (Age vs Days since last interaction)
- **Null Hypothesis H_0:** There is no relationship between `Age` and `DaysSinceLastInteraction`. ($r = 0$)
- **Alternative Hypothesis H1:** There is a significant positive relationship between `Age` and `DaysSinceLastInteraction` $r > 0$.

4. Does state-wise spend vary significantly?

- **Statistical Test:** One-Way ANOVA
- **Null Hypothesis H_0:** The mean `MonthlySpend` is identical across all states. ($\mu(\text{State A}) = \mu(\text{State B}) = \dots$)
- **Alternative Hypothesis H1:** At least one state has a significantly different mean `MonthlySpend` compared to the others.

Step 6: Run Hypothesis Tests

Business Purpose: Validate or reject your assumptions with confidence.

- Define null and alternate hypotheses
- Define null and alternate hypotheses
- Choose test based on data types
- Check assumptions: normality, independence, homogeneity of variance
- Interpret p-values and confidence intervals

Do males and females spend differently?

```
male = US_Customer_Insights_Dataset[US_Customer_Insights_Dataset['Gender'] == 'Male']['MonthlySpend']
female = US_Customer_Insights_Dataset[US_Customer_Insights_Dataset['Gender'] == 'Female']['MonthlySpend']

result = (ttest_ind(male,female))
p_value = result.pvalue
print(f'p value {p_value}')

if p_value < 0.05:
    print("H0 Rejected and H1 Accepted")
else:
    print("H0 Accepted")
```

```
p value 0.7344892727022861
H0 Accepted
```

- **Hypothesis Decision:** Fail to reject the Null Hypothesis H0.
- **Statistical Finding:** The test confirms that the Null Hypothesis H0 stands. There is no statistically significant difference between the mean `MonthlySpend` of male and female customers ($p > 0.05$).
- **Business Insight:** Gender does not act as a driving factor for customer purchasing volume in this dataset. Marketing campaigns and budget allocations targeting spend habits should treat male and female segments equally, as their spending distributions are statistically identical.

Does education level impact average monthllyspend?

```

high_school = US_Customer_Insights_Dataset[US_Customer_Insights_Dataset['Education'] == 'High School']['MonthlySpend']
master = US_Customer_Insights_Dataset[US_Customer_Insights_Dataset['Education'] == 'Master']['MonthlySpend']
phd = US_Customer_Insights_Dataset[US_Customer_Insights_Dataset['Education'] == 'PhD']['MonthlySpend']
bachelor = US_Customer_Insights_Dataset[US_Customer_Insights_Dataset['Education'] == 'Bachelor']['MonthlySpend']
associate = US_Customer_Insights_Dataset[US_Customer_Insights_Dataset['Education'] == 'Associate']['MonthlySpend']

f_stat, p_value = f_oneway(high_school, master, phd, bachelor, associate)

print(f"F-Statistic: {f_stat:.4f}")
print(f"P-value: {p_value}")

if p_value < 0.05:
    print("H0 Rejected and H1 Accepted")
else:
    print("Fail to reject H0 (The Null Hypothesis Stands)")

```

```

F-Statistic: 0.2288
P-value: 0.9223594677600564
Fail to reject H0 (The Null Hypothesis Stands)

```

- **Hypothesis Decision:** Fail to reject the Null Hypothesis H0.
- **Statistical Finding:** The One-Way ANOVA test confirms that the Null Hypothesis H0 stands ($F(4) = 0.2288$, $p = 0.9224$). There is no statistically significant difference in the mean `MonthlySpend` across the different education levels (High School, Associate, Bachelor, Master, PhD).
- **Business Insight:** A customer's formal level of education does not drive or predict their monthly purchasing volume in this market segment. The exceptionally high p-value (92.24%) emphasizes that any minor variance observed between the groups is entirely due to random sampling fluctuations.
- **Strategic Recommendation:** Marketing strategies, product pricing tiers, and promotional campaigns do not need to be segmented or tailored based on educational degrees, as spending behavior remains uniform across all educational backgrounds.

Are older people less active?

```

corr = US_Customer_Insights_Dataset[['Age', 'DaysSinceLastInteraction']].corr()
corr

```

	Age	DaysSinceLastInteraction
Age	1.00000	-0.00397
DaysSinceLastInteraction	-0.00397	1.00000

- **Hypothesis Decision:** Fail to reject the Null Hypothesis H_0 .
- **Statistical Finding:** The Pearson correlation test indicates that the Null Hypothesis H_0 stands. The correlation coefficient between `Age` and `DaysSinceLastInteraction` is practically zero ($r = -0.0040$). There is no linear relationship between a customer's age and their interaction patterns.
- **Business Insight:** Older individuals are not less active in this customer ecosystem. Customer engagement—measured by the number of days since their last interaction—remains completely uniform across all generations, meaning age is a poor metric for predicting user inactivity or platform churn.
- **Strategic Recommendation:** Do not build re-engagement campaigns or retention flows around customer age brackets. Instead, look into other potential churn indicators (such as historical spending volume or customer support tickets) to address drop-offs in platform activity

Does state-wise spend vary significantly?

```
state_groups = [
    group["MonthlySpend"].values
    for name, group in US_Customer_Insights_Dataset.groupby("State")
]

f_stat, p_value = f_oneway(*state_groups)

print(f"F-Statistic: {f_stat:.4f}")
print(f"P-value: {p_value}")

if p_value < 0.05:
    print("H0 Rejected and H1 Accepted")
else:
    print("Fail to reject H0 (The Null Hypothesis Stands)")
```

```
F-Statistic: 1.1178
P-value: 0.34571886479464203
Fail to reject H0 (The Null Hypothesis Stands)
```

- **Hypothesis Decision:** Fail to reject the Null Hypothesis (H_0).
- **Statistical Finding:** The One-Way ANOVA test confirms that the Null Hypothesis (H_0) stands ($F = 1.1178$, $p = 0.3457$). There is no statistically significant difference in the mean `MonthlySpend` across the different US states in the dataset.
- **Business Insight:** Customer value and purchasing power are regionally uniform. A customer's geographic location (state) does not drive or predict their monthly spending behavior. The p-value (34.57%) is well above our significance threshold ($\alpha = 0.05$), proving that any slight variations in average spending between states are entirely due to random sampling fluctuations.
- **Strategic Recommendation:** Regional or state-level custom pricing tiers and specialized regional marketing distributions are unnecessary for driving raw sales volume. Marketing assets can safely remain generalized at a national scale, saving operational overhead and design resources.

Step 7: Present Business Insights

1. Majority of spending is between 0- 750\$ as can be seen in th box plot. Thus there can be an increase in products within that cost range
2. There is a small difference beyween the monthly spending of married and unmarried people. Where unmarried people tend to spend more than the married. Thus more promotional messages to unmarried people can increase sales
3. Our independent t-test yielded a p-value of 0.7345, meaning there is no statistical variance in spending patterns between male and female cohorts. Thus we can avoid costly gender-segmented advertising or separate product lines. Marketing assets and promotional campaigns should stay broad, inclusive, and standardized across all genders.
4. The state-by-state One-Way ANOVA test showed a p-value of 0.3457, meaning geographical location does not dictate spending habits. Consolidate localized marketing budgets. Instead of managing ten separate localized state campaigns, launch a singular, high-budget national campaign to reduce administrative workloads and design costs while maximizing reach.
5. The correlation between customer age and days since last interaction was practically non-existent ($r = -0.0040$). Disregard age boundaries when deploying automated win-back emails or churn-prevention alerts. Treat a drop-off in customer interaction identically whether the user is 22 or 65, utilizing identical timing triggers for all churn-prevention automation.

